Components required in your mini-project

Choose one tool per layer. Three layers are mandatory (marked required).

#	Component	Function	Tool options (pick one)	Required
1	**Data source** The event stream	Timestamped data arriving continuously	CSV/JSONL replay, IoT feed, API poller, logs	Any source
2	**Streaming tool** Transports the stream	Buffering, ordering, windowing of events	Kafka, Storm, Spark, Flink, Kinesis	**Required**
3	**Stream processor** Computes live metrics	Rolling avg, peak, running totals, state	Python, Spark, Flink, Kafka Streams, Faust	May reuse #2
4	**ML algorithm** Scores each event	Anomaly detection, classify or forecast	IsolationForest, SVM, K-Means, ARIMA	**Required**
5	**Database** Stores state + history	Serving layer the dashboard queries	MySQL, PostgreSQL, MongoDB, InfluxDB	Any store
6	**Visualization** Live auto-refresh view	Presents results, updates on its own	Streamlit, Dash, Grafana, Power BI	**Required**

Pipeline flow:

Data source → Streaming tool → Processor → ML → Database → Live dashboard